Algebraic Topology: a beginner's course

This lecture:

AlgTop13: More applications of winding numbers


Lecture 13 More applications of winding numbers

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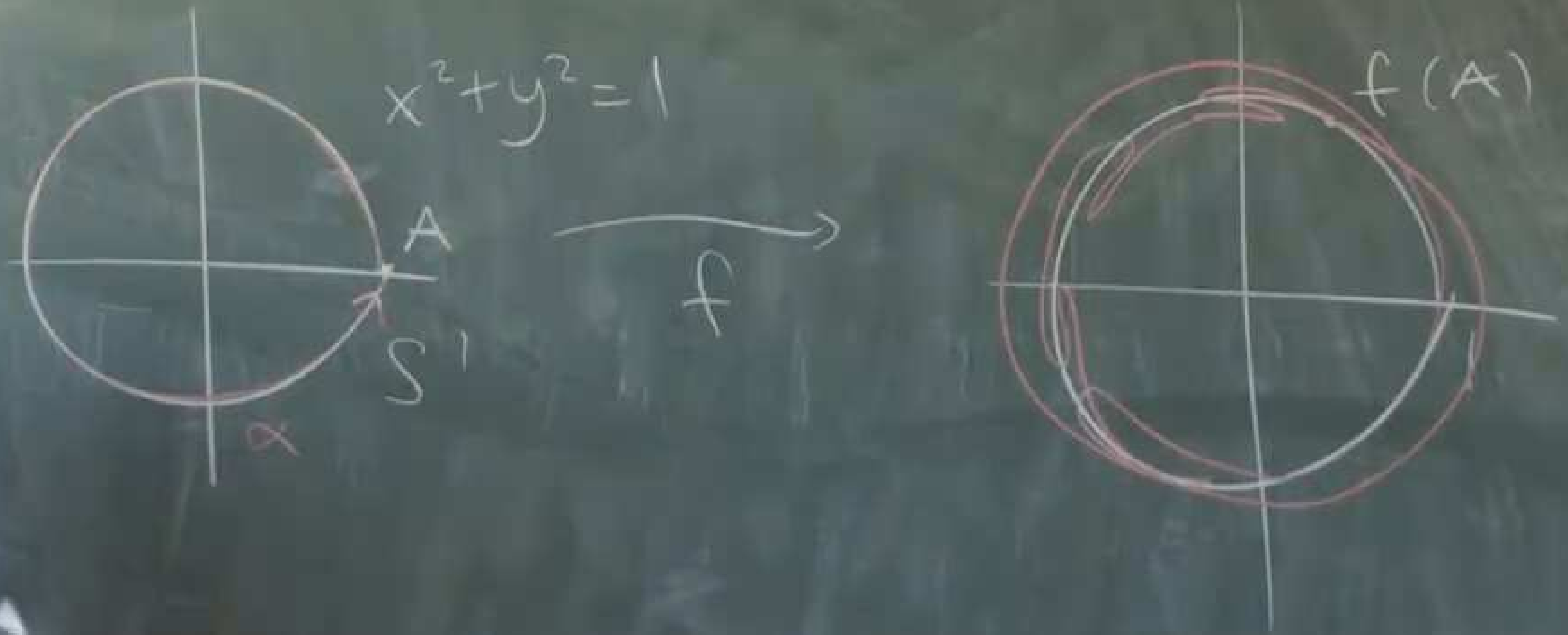
$$f: S^1 \longrightarrow S^1$$



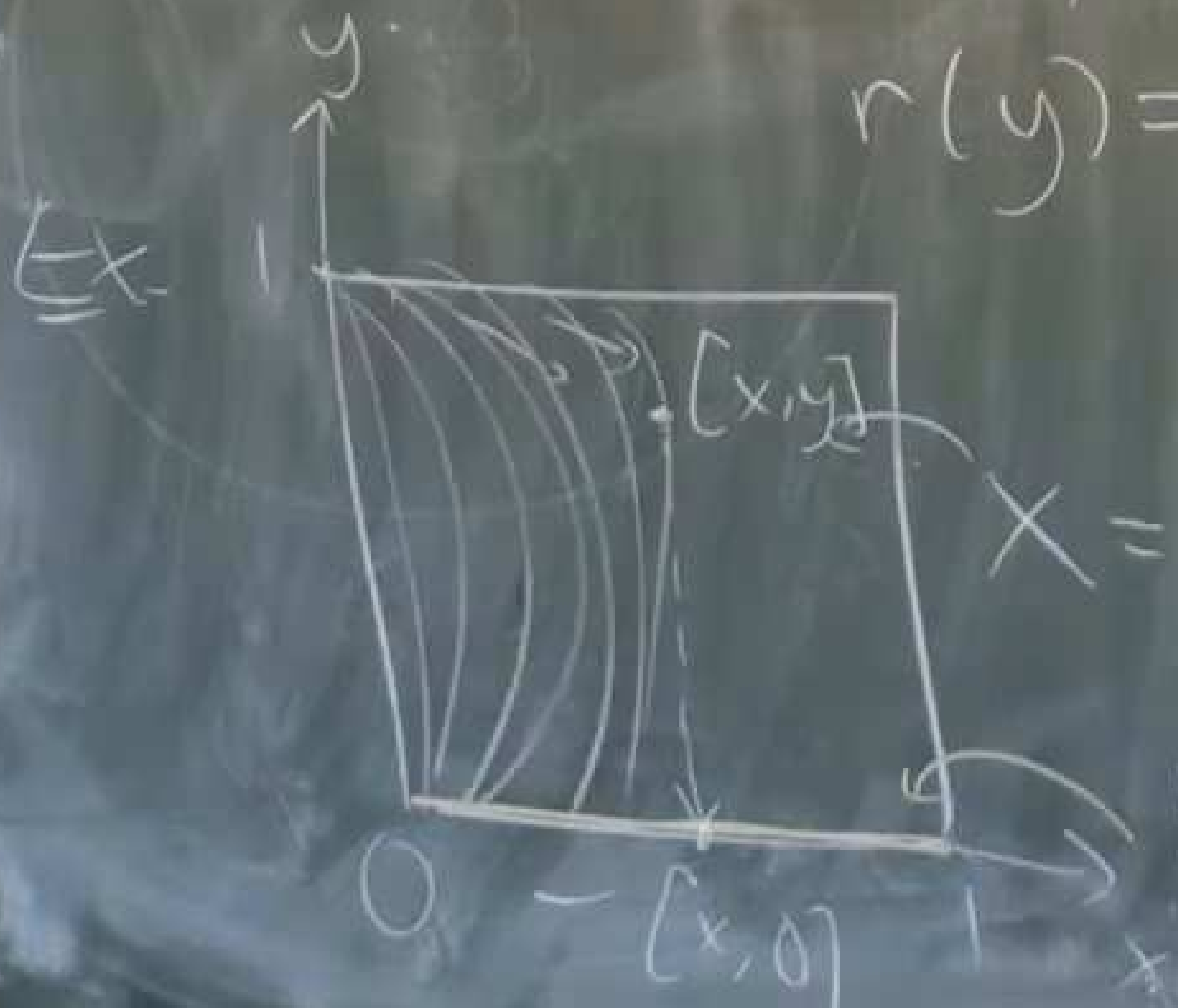
Retraction If X, Y top. spaces, Y contained in X,

a **retraction** from X to Y, $r: X \to Y$ is

a continuous map such that

$r(y) = y$ for

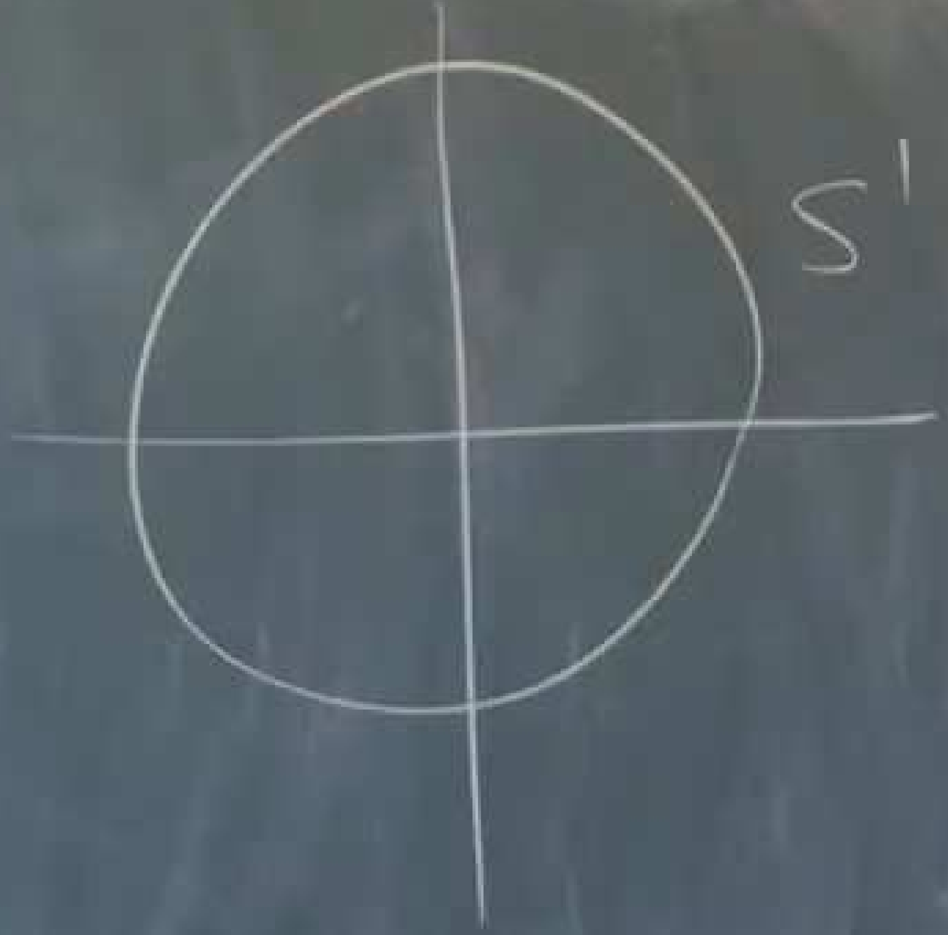
Ex.



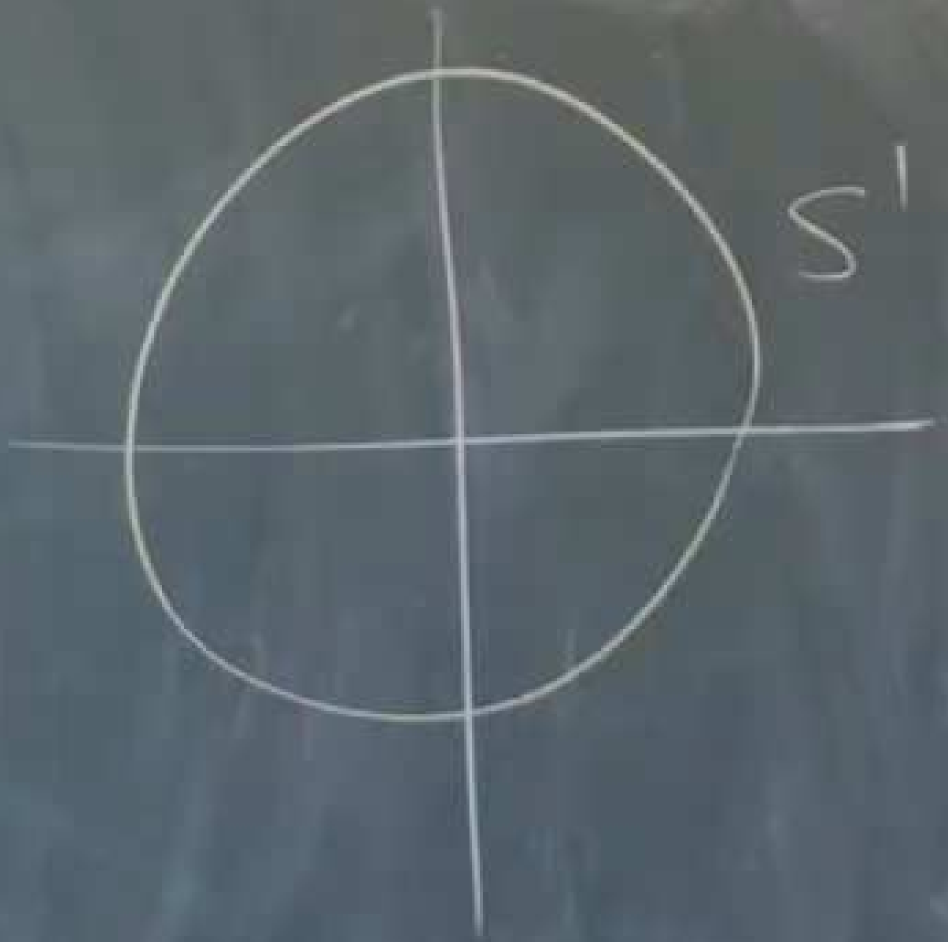
$X = I \times I$ ... 0]

$Y = I$ ...

Proof Suppose $r: D \to S^1$ is a retraction.



Proof Suppose $r: D \to S^1$ is a retraction.



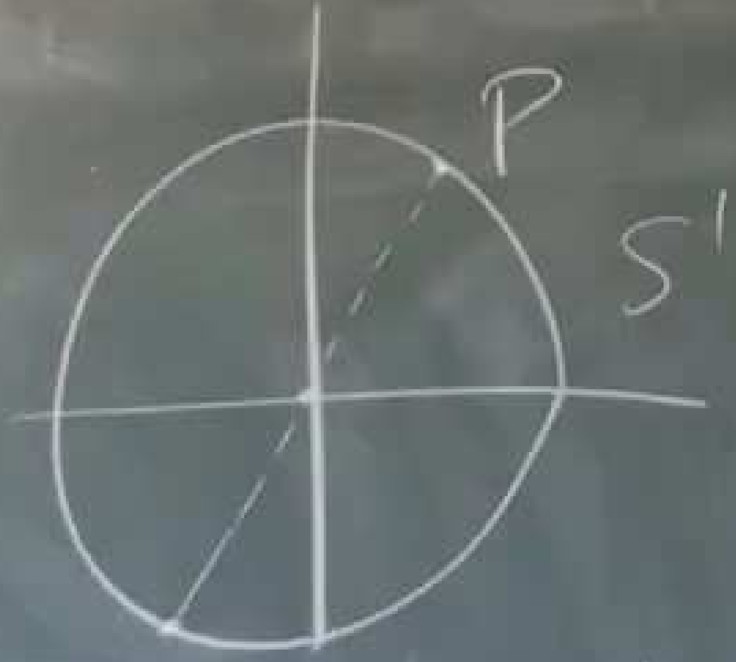
So this gives a continuous family between

p_0 - deg 1 map and p_1 - deg 0 map (from S^1 to S^1)

But degree cannot change since p_t is continuous

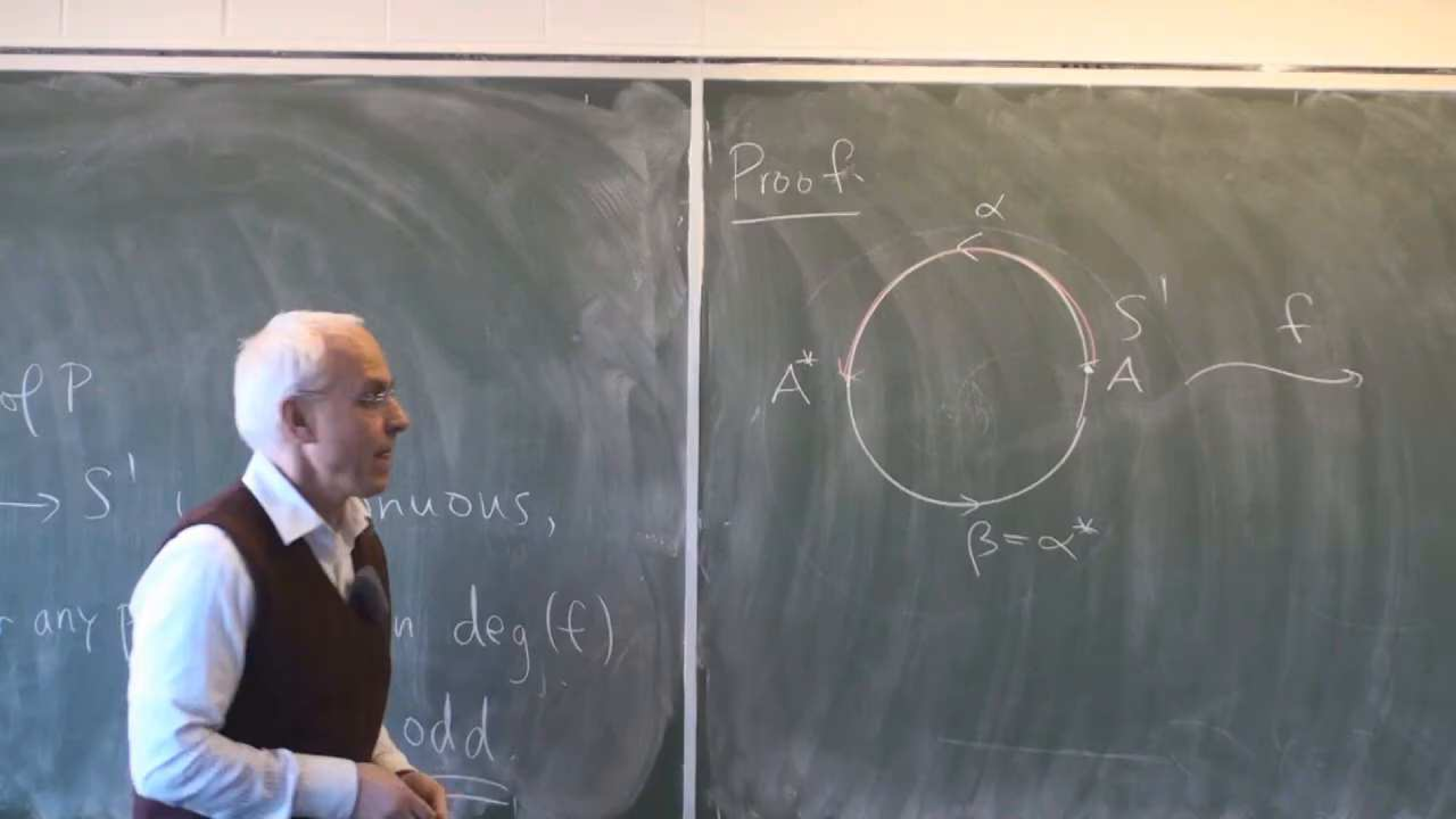
X So a retraction can't exist. //

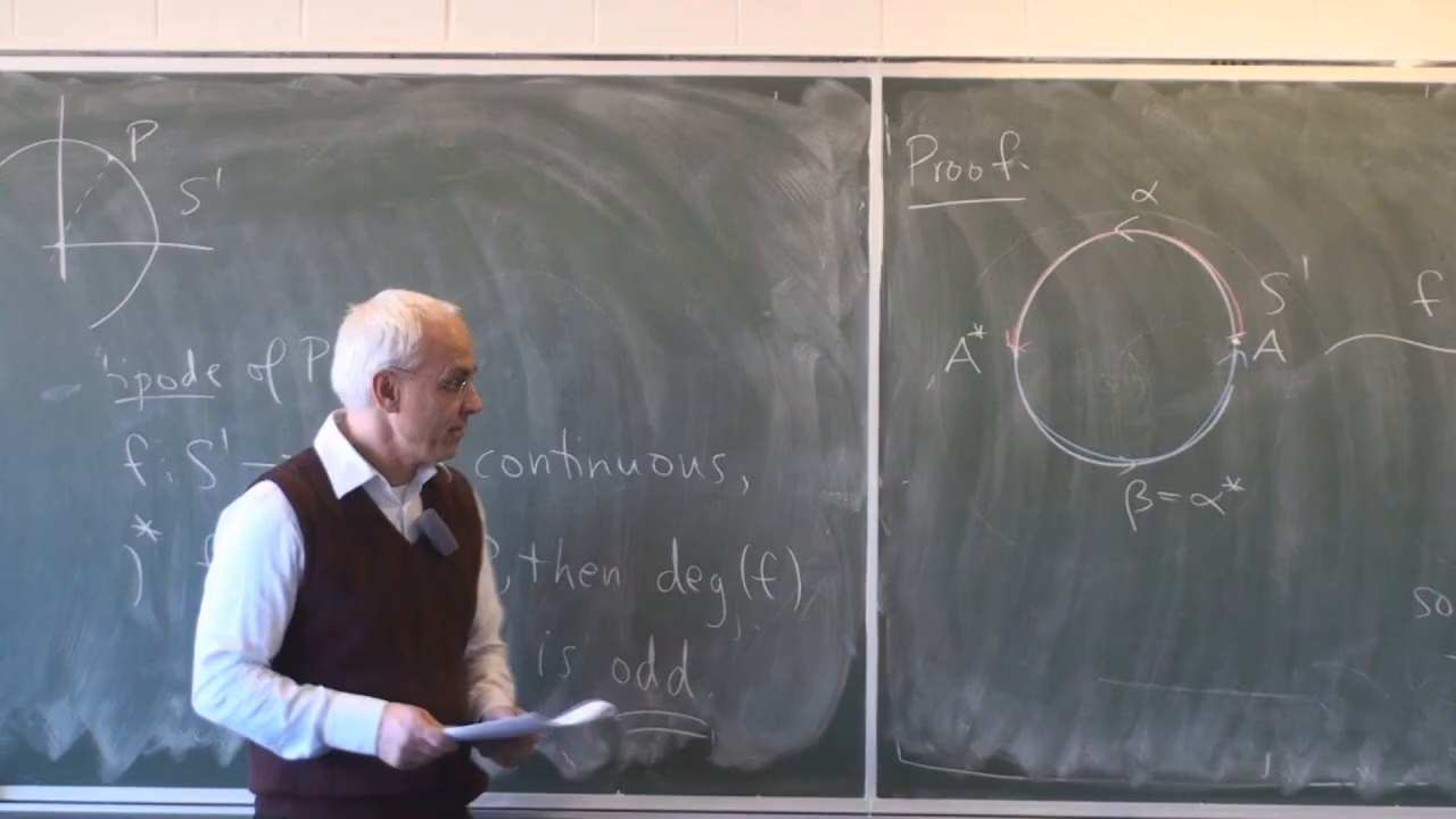
Antipodes



P^* antipode of P

Lemma (Borsuk) If $f: S^1 \to S^1$ is continuous, and $f(P^*) = f(P)^*$ for any point P, then $\deg(f)$ is odd.


of P
→ S¹ nuous,
r any n deg(f)
odd.
Proof.
α
S¹
f
A*
A
β=α*


P
S'
ipode of P
f: S'
continuous,
then deg(f)
is odd
Proof.
α
S'
A*
A
β=α*
f
so

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next in the series:

AlgTop14: The Ham sandwich theorem and the continuum